CLINICAL · NOSE & SINUSES

# Chronic rhinosinusitis and nasal polyps

Chronic rhinosinusitis is a long-standing inflammation of the nose and sinuses that lasts beyond twelve weeks, with or without polyps. It is not simply an infection that refuses to clear: anatomy, cilia, allergy, asthma, biofilms and the patient's own immune pattern all play their part. Most patients are controlled with saline and steroid sprays used properly; surgery opens the sinuses so that the medicine can reach them.

AT A GLANCE

- 12 weeks of symptoms with objective findings.
- Saline and topical steroids first.
- Surgery helps medical treatment reach the mucosa.
- Children with polyps need a CF test.
- EPOS 2020: primary localised or diffuse, type 2 or not; biologics for recurrent type 2 polyps meeting 3 of 5 criteria.

## 1 CHAPTER ONE Overview

Inflammation beyond twelve weeks.

### 1.1 Definition

Inflammation of the nose and sinuses lasting 12 weeks or more, with at least two cardinal symptoms, one of which must be blockage or discharge, plus objective evidence on endoscopy or CT.

#### 1.1.1 Cardinal symptoms

- Nasal blockage or congestion.
- Anterior or posterior (postnasal) discharge, often purulent.
- Facial pain or pressure.
- Reduced or lost sense of smell.

#### 1.1.2 Objective signs

> Endoscopy showing oedema or polyps in the middle meatus, or mucopus there; or CT showing mucosal changes in the ostiomeatal complex or sinuses.

### 1.2 Two phenotypes

- **CRS without nasal polyps:** more often infective and neutrophilic.
- **CRS with nasal polyps:** usually type 2 (eosinophilic) inflammation driven by IL-4, IL-5 and IL-13, often with asthma.

### 1.3 Acute exacerbation

A sudden worsening that returns to the chronic baseline after treatment.

## 2 CHAPTER TWO Causes and pathogenesis

A vicious cycle.

### 2.1 Predisposing factors

#### 2.1.1 Anatomical

Structural variants that narrow the ostiomeatal complex: septal deviation, *concha bullosa* (a pneumatised middle turbinate), a paradoxical middle turbinate, Haller (infraorbital ethmoid) cells, prominent agger nasi cells, and a pneumatised or medially bent uncinate process.

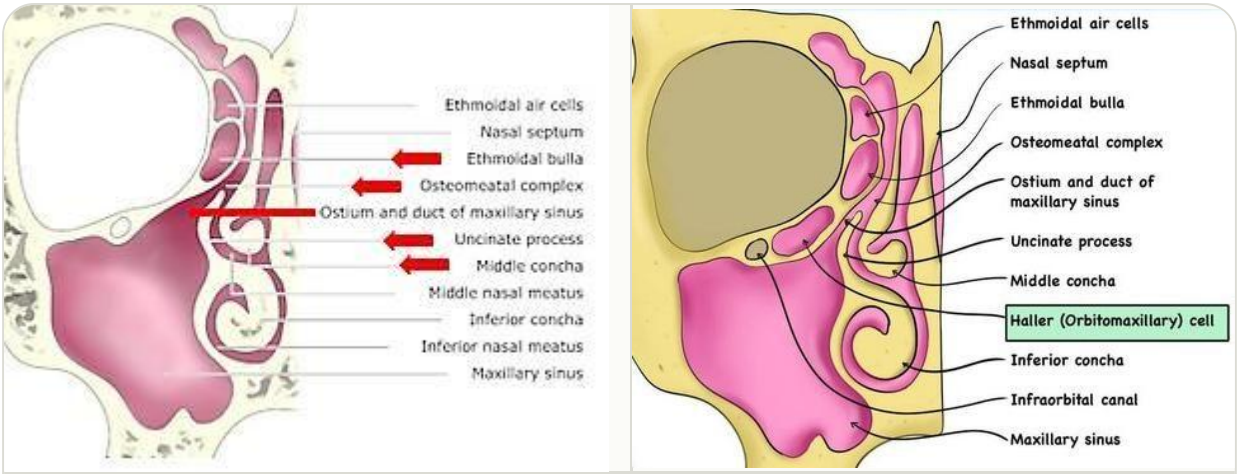



**FIGURE 1** The ostiomeatal complex in coronal section, and a Haller cell narrowing the maxillary outflow.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 146

### 2.1.2 Mucociliary failure

Infection, smoking and pollution impair cilia; in *primary ciliary dyskinesia* (Kartagener syndrome with situs inversus and bronchiectasis) cilia do not beat; in *cystic fibrosis* the mucus is too thick to move; *Young syndrome* combines sinopulmonary disease with azoospermia.

### 2.1.3 Host and environment

Allergy, asthma (about half of CRS patients), aspirin sensitivity, immune deficiency (IgG deficiency, HIV, chemotherapy, transplant drugs), smoking, occupational dusts and fumes, dental infection, and osteitis of the sinus bone after surgery.

### 2.1.4 Biofilms

Colonies of bacteria in a protective polysaccharide matrix that resists antibiotics and keeps the disease going.

## 2.2 Organisms

Polymicrobial: *Staphylococcus aureus* (sometimes methicillin-resistant), *Pseudomonas*, *Klebsiella*, *E. coli* and anaerobes, unlike the organisms of acute disease.

## 2.3 The cycle

Ostial obstruction leads to stasis, low oxygen and infection; infection and inflammation damage cilia and swell the mucosa further, which blocks the ostium more.

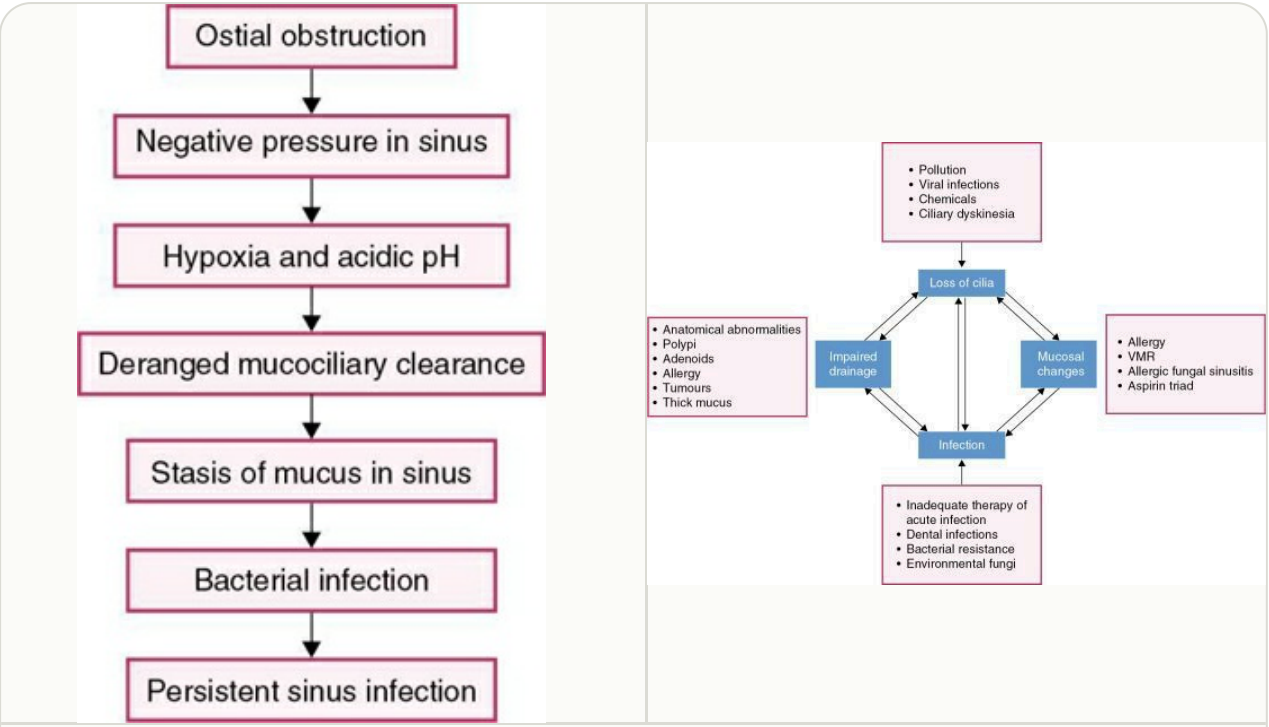



**FIGURE 2** First, the sequence of events after sinus ostial obstruction. Second, the factors in the vicious cycle of chronic sinusitis.

Dhingra, figs 37.1 and 37.2

# 3 CHAPTER THREE Nasal polyps

Grapes in the nose.

## 3.1 Two kinds

### ANTROCHOANAL OR ETHMOIDAL POLYPS?

| | ANTROCHOANAL POLYP | ETHMOIDAL POLYPS |
|---|---|---|
| **Age** | Children and young adults | Adults |
| **Cause** | Infection | Allergy, multifactorial inflammation |

| | ANTROCHOANAL POLYP | ETHMOIDAL POLYPS |
|---|---|---|
| **Number and side** | Single, one-sided | Multiple, both sides |
| **Origin** | Maxillary sinus, near the accessory ostium | Ethmoid cells, uncinate, middle turbinate, middle meatus |
| **Growth** | Backwards to the choana; may hang behind the soft palate | Forwards; may appear at the nostril |
| **Shape** | Three parts: antral, choanal and nasal | Small, grape-like masses |
| **Recurrence** | Uncommon after complete removal | Common |

## 3.2 Antrochoanal polyp

Also called *Killian's polyp*. It arises near the accessory ostium of the maxillary sinus and has three parts: a thin antral stalk, a round choanal part, and a flattened nasal part. It causes one-sided blockage that may become bilateral when the choanal part fills the nasopharynx, with a hyponasal voice. It can be missed on anterior rhinoscopy and is seen with an endoscope or hanging behind the soft palate. CT shows an opaque antrum and the mass passing through the ostium.

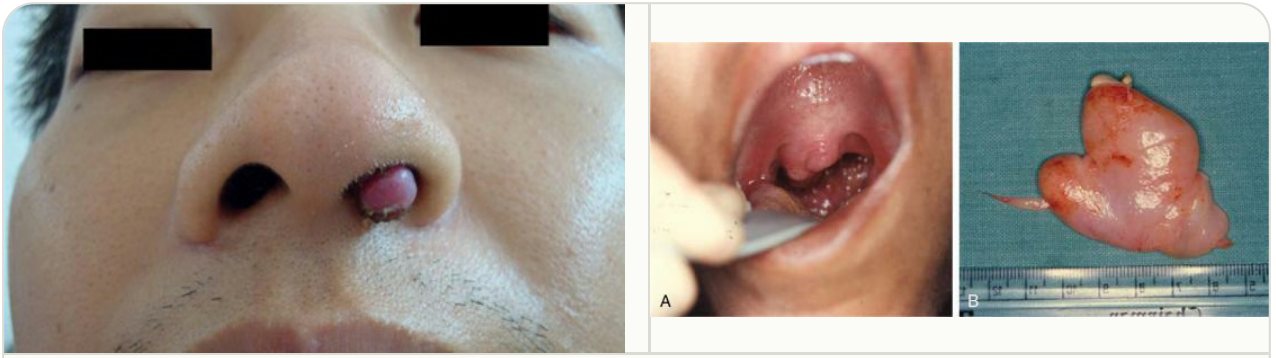



**FIGURE 3** First, an antrochoanal polyp at the left nostril of a 14-year-old. Second, one hanging into the oropharynx behind the soft palate, and the polyp after removal.
Dhingra, figs 32.1 and 32.2

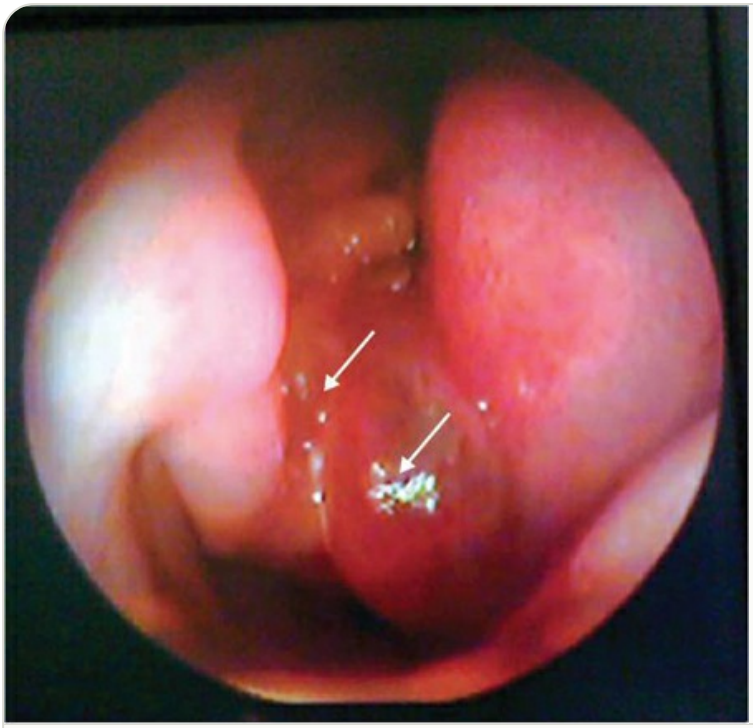


**FIGURE 4** Endoscopic view of a choanal polyp.
Dhingra, fig. 32.3

#### 3.2.1 Treatment

Endoscopic removal including its antral attachment, which prevents recurrence. Caldwell-Luc surgery is avoided, especially in children, because it damages the developing teeth.

## 3.3 Ethmoidal (bilateral) polyps

#### 3.3.1 Associations

- Chronic rhinosinusitis, allergic and non-allergic, and NARES.
- Asthma (about 7% of asthmatics have polyps).
- Aspirin intolerance: about a third; *Samter's triad* (aspirin-exacerbated respiratory disease) is polyps, asthma and aspirin sensitivity.
- Cystic fibrosis (about 20%); polyps in a child should prompt a sweat test.
- Allergic fungal sinusitis (almost all).
- Kartagener and Young syndromes, Churg-Strauss syndrome (eosinophilic granulomatosis with polyangiitis), and nasal mastocytosis.

#### 3.3.2 Pathology

Oedematous mucosa of the middle meatus and turbinate prolapses into the nose; sessile polyps become pedunculated with gravity and sneezing. Large fluid-filled spaces and eosinophils lie in the stroma; the surface may become squamous where exposed.

### 3.3.3 Site

Always from the lateral wall, usually the middle meatus (uncinate, bulla, sinus ostia, middle turbinate); allergic polyps almost never arise from the septum or floor.

### 3.3.4 Clinical picture

Progressive bilateral blockage, loss of smell, headache from sinusitis, sneezing and watery discharge. Polyps are smooth, glistening, pale, grape-like, insensitive to probing and do not bleed. Long-standing polyposis broadens the nose and widens the intercanthal distance (the *frog face*, or Woakes syndrome in its destructive form).

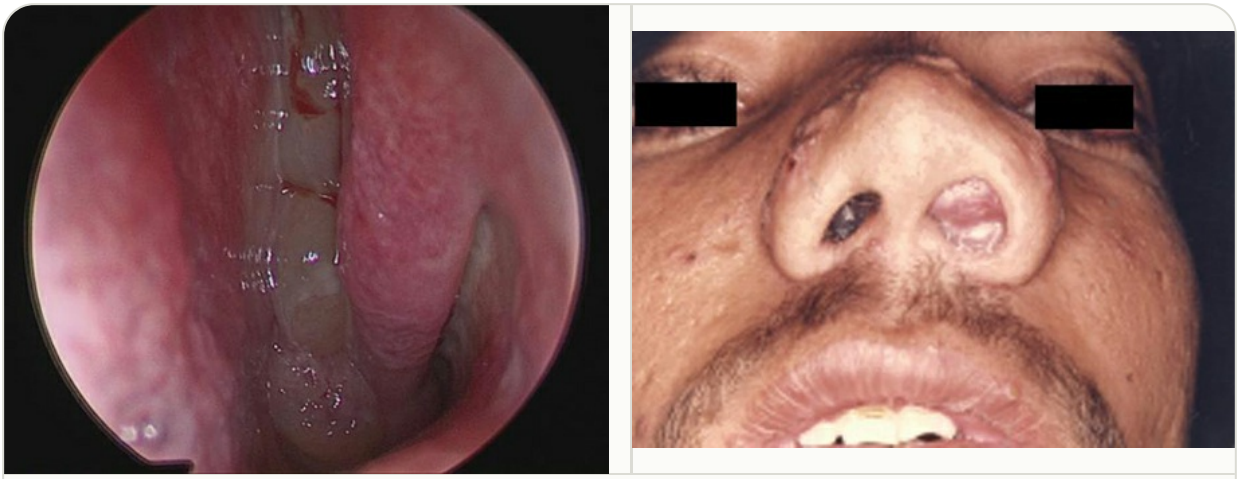


**FIGURE 5** First, multiple polyps on endoscopy. Second, a polyp protruding from the left nostril. Dhingra, figs 32.4 and 32.5

### 3.3.5 Differential diagnosis

A blob of mucus (disappears on blowing), a hypertrophied middle turbinate (pink and bony on probing), angiofibroma (bleeds, firm), inverted papilloma and malignancy (one-sided, fleshy, friable).

**ONE-SIDED POLYP IN AN ADULT**

> A unilateral polyp is not simple polyposis until histology says so: inverted papilloma, carcinoma, esthesioneuroblastoma and encephalocele all present this way.

# 4 CHAPTER FOUR Assessment

Endoscopy and CT.

## 4.1 Nasal endoscopy

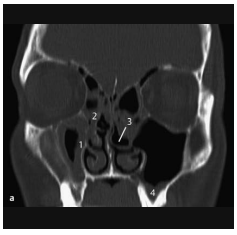
Before and after decongestion: oedema, polyps and pus in the middle meatus and sphenoethmoidal recess. Polyps are graded by extent (for example the Lund-Kennedy score).

## 4.2 CT of the sinuses

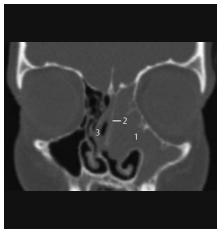
Non-contrast CT in three planes is the gold standard: it shows the extent of disease (scored by the Lund-Mackay system), anatomical variants and bony erosion, and maps the anatomy for surgery. Not for diagnosis alone, but before any operation.

**IMAGING IN THE APP** 5 cases

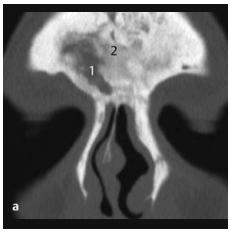
Nasal and unilateral polyps, cystic fibrosis and mucoceles.



**Nasal Polyps**
Infection & inflammation · 10 images



**Unilateral Nasal Polyps**
Infection & inflammation · 1 image



**Cystic Fibrosis**
Infection & inflammation · 3 images



**Mucocele of the Frontal Recess**
Infection & inflammation · 3 images



**Mucocele**
Infection & inflammation · 13 images

### 4.3 Other tests

Allergy testing, a sweat test or genetics for cystic fibrosis in children, ciliary studies, immunoglobulins, eosinophil count and ANCA when vasculitis is suspected, and culture from the middle meatus.

### 4.4 Symptom scores

Patient-reported scores such as SNOT-22 measure the burden and the response to treatment.

## 5 CHAPTER FIVE Medical treatment

Saline and steroids first.

### 5.1 CRS without polyps

- **Saline irrigation:** high-volume rinses clear mucus, bacteria and allergens and disrupt biofilms.
- **Intranasal corticosteroids:** sprays or drops; better after decongestion and with correct technique.
- **Antibiotics:** culture-guided for exacerbations; long-term low-dose macrolides (over 12 weeks) for their anti-inflammatory effect in selected patients.
- **Treat allergy:** antihistamines, leukotriene antagonists, immunotherapy.

### 5.2 CRS with polyps

- **Steroid sprays or drops,** long-term.
- **Short oral steroid courses** shrink polyps for a few weeks; limit their number because of side effects.
- **Doxycycline** for a few weeks reduces polyp size modestly.
- **Treat asthma** with the respiratory team.
- **Aspirin-exacerbated disease:** leukotriene modifiers, and aspirin desensitisation after surgery.

- **Biologics:** monoclonal antibodies against type 2 inflammation (anti-IL-4 receptor, anti-IgE, anti-IL-5) for severe, recurrent polyposis despite surgery and steroids, especially with asthma.

# 6 CHAPTER SIX Surgery

Opening the sinuses for the medicine.

## 6.1 Endoscopic sinus surgery

For patients whose symptoms persist despite adequate medical treatment. It removes polyps and diseased tissue, opens the natural ostia, corrects obstructing anatomy, and makes a cavity that saline and steroids can reach. Medical treatment continues afterwards. Revision is common in polyposis. In cystic fibrosis and ciliary disease, wide dependent drainage of the maxillary and sphenoid sinuses is made.

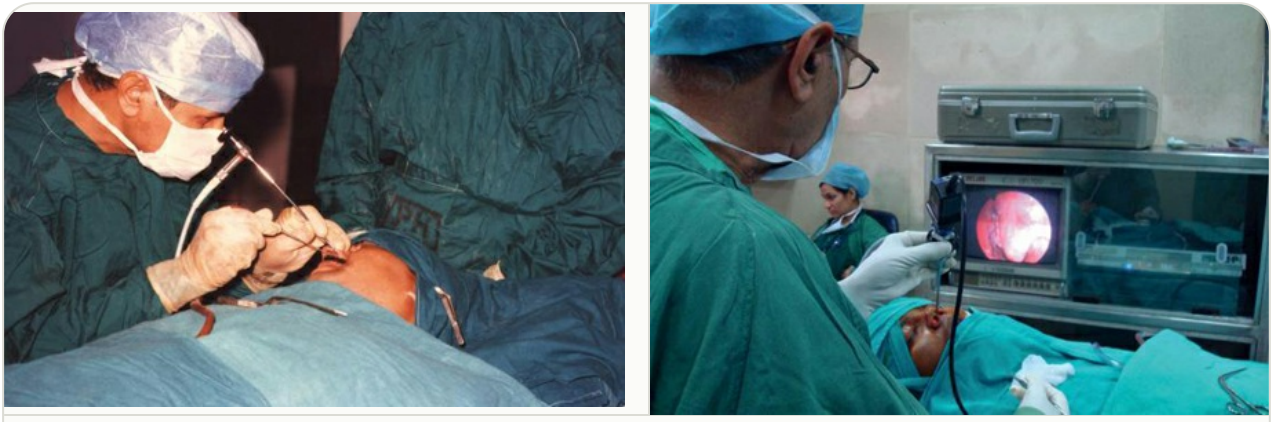


**FIGURE 6** Functional endoscopic sinus surgery, without and with a camera and monitor.
Dhingra, figs 37.4 and 37.5

## 6.2 Historical operations

Intranasal polypectomy alone, Caldwell-Luc operation, intranasal and external ethmoidectomy, and external frontoethmoidectomy (Howarth or Lynch); now largely replaced by endoscopic surgery. See the nose and sinus operations topic.

**OPERATIVE GUIDES IN THEATRE** 5 operations

**Antral Lavage; Endoscopic Middle Meatal Antrostomy/Antrochoanal Polyp**
Antrostomy
**Endoscopic Ethmoidectomy**
Ethmoidectomy
**Endoscopic Sphenoidotomy**
Sphenoidotomy
**Endoscopic Approach to the Frontal Sinus**
Frontal approach
**Draf III, Endoscopic Modified Lothrop**
Draf III

CHAPTER SEVEN

# 7 Chronic sinusitis by sinus

Old names, still examined.

## 7.1 Chronic maxillary sinusitis

Purulent discharge, postnasal drip, cacosmia, often dental in origin; endoscopic middle meatal antrostomy has replaced antral washouts, inferior meatal antrostomy and Caldwell-Luc operations.

## 7.2 Chronic frontal sinusitis

Frontal headache and discharge; endoscopic frontal sinusotomy, extended to a Draf III (modified Lothrop) procedure when needed; external frontoethmoidectomy and osteoplastic obliteration are reserved for failures.

## 7.3 Chronic ethmoiditis and sphenoiditis

Usually part of pansinusitis; endoscopic ethmoidectomy and sphenoidotomy.

EPOS 2020: ENDOTYPES, IMAGING AND BIOLOGICS (OXFORD TEXTBOOK)

**Classification.** Whether polyps are present says little about the disease underneath, so EPOS 2020 classifies chronic rhinosinusitis by mechanism. *Primary* disease is split into **localised** (one side or one sinus system, such as isolated sphenoid disease or allergic fungal rhinosinusitis) and **diffuse** (both sides), and each into **type 2** (eosinophilic) and **non-type 2** endotypes. *Secondary* disease arises from something else: odontogenic sinusitis, a tumour, immunodeficiency, primary ciliary dyskinesia,

cystic fibrosis or eosinophilic granulomatosis with polyangiitis. Localised non-type 2 disease has the best prognosis; diffuse type 2 disease is part of a wider airway disorder and tends to recur.

**Central compartment atopic disease** is a pattern of allergic inflammation centred on the middle turbinate head, superior septum and posterosuperior nasal cavity, in contrast to the true polyps from the middle meatus seen in eosinophilic disease. Thick "chewing gum" eosinophilic mucin suggests eosinophilic chronic rhinosinusitis or allergic fungal disease. A blood eosinophil count above $0.25 \times 10^9$/L supports a type 2 endotype, and above $0.45 \times 10^9$/L predicts recurrence.

**Imaging.** Incidental changes are common: mucous retention cysts in 10 to 15% and mucosal thickening in a quarter of normal people, who score up to about 4 on Lund–Mackay; EPOS 2020 takes a score above 5 as strongly suggestive of disease. MRI is too sensitive to diagnose chronic rhinosinusitis, showing thickening in about half the general population.

**Biologics.** Monoclonal antibodies against type 2 inflammation (dupilumab against the IL-4 receptor, omalizumab against IgE, and mepolizumab against IL-5) are indicated by EPOS 2020 for bilateral polyps after sinus surgery (or in patients unfit for it) with at least three of: evidence of type 2 disease (tissue eosinophils ≥10 per high-power field, blood eosinophils $\geq 0.25 \times 10^9$/L or total IgE ≥100), need for systemic steroids (two or more courses a year or long-term use), SNOT-22 of 40 or more, anosmia on testing, and asthma needing inhaled steroids. Response is reviewed at 16 weeks and one year.

**Surgery** for diffuse type 2 disease aims for a complete, wide "neo-sinus" cavity that gives topical steroids access, rather than a limited opening of ostia, which suits localised disease. Long-term macrolides help selected non-type 2 patients with normal IgE.

# 8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

## 8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition** and diagnostic criteria.
- **Predisposing factors,** especially anatomical variants and ciliary disorders.

- **Antrochoanal versus ethmoidal polyps.**
- **Samter's triad,** and polyps in a child (cystic fibrosis).
- **Medical treatment,** including biologics.
- **Indications for endoscopic sinus surgery.**

## 8.2 Common pitfalls

COMMON PITFALLS

- Repeated antibiotics without steroids or saline.
- Calling a one-sided polyp benign without histology.
- Operating before medical treatment has been tried properly.
- Forgetting the sweat test in a child with polyps.

Written for this app from Oxford Textbook of Otolaryngology ch. 37, 38, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 32, 37 and K. J. Lee ch. 29. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

## Nasal Polyps

### IMAGING FINDINGS

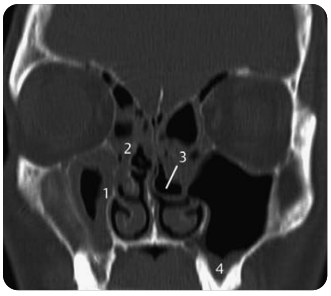
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

### TOP DIFFERENTIALS

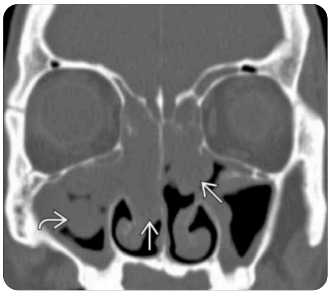
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

### CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist




CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis

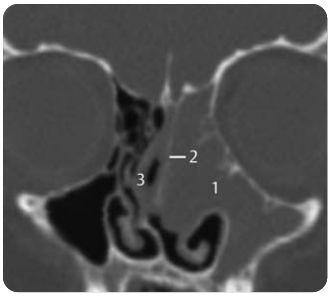
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

## Unilateral Nasal Polyps

- Destruction of bony outlines (with infiltrative characteristics) is suggestive of inverted papilloma, malignancies, or fungal infection.
- Calcifications in the opacified area are suggestive of fungal infection. Resorption of bone around tooth roots suggests odontogenic infection.
- Beware of pathology in the sphenoethmoidal fossa (i.e., juvenile angiofibroma).

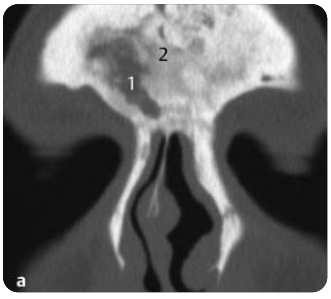



CT, coronal. Complete opacification of the left maxillary sinus and ethmoid (1) is seen. Medial displacement of the middle turbinate (2) and nasal septu…

PARANASAL SINUSES · INFECTION & INFLAMMATION

## Cystic Fibrosis

- Destruction of bony outlines with infiltrative signs is more characteristic of malignancy.
- Expansion is more likely to be due to benign processes, such as polyposis or mucocele.
- Calcifications in the opacified area are suggestive of fungal infections.
- Resorption of bone around tooth roots indicates dental pulp infections.
- In children, expansion due to polyps can result in cosmetic problems of the exterior aspect of the nose.




CT, coronal. There is opacification of the right frontal sinus (1) with surrounding osteitic changes (2) such as increased thickness and sclerosis…

PARANASAL SINUSES · INFECTION & INFLAMMATION

## Mucocele of the Frontal Recess

- Beware of large osseous lesions of the skull base, removal of which may result in protrusion of intracranial contents or marsupialization of the mucocele.
- In meningo(encephalo)cele or intracranial tumors, persistent leakage of cerebrospinal fluid may occur at biopsy.




CT with intravenous contrast enhancement, axial. The lesion in the anterior skull base (1) in the area of the frontal recess shows signs of…

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

## Mucocele

### IMAGING FINDINGS

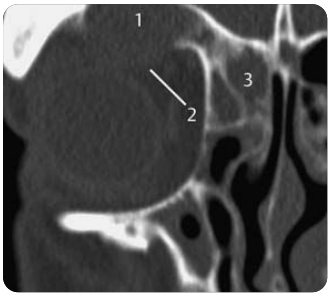
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

### TOP DIFFERENTIALS

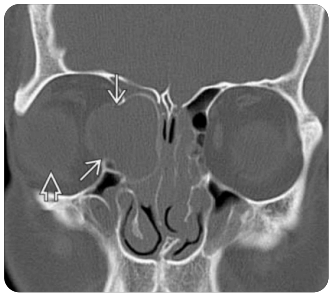
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

### CHECKLIST

- Orbit and skull base erosion
- Frontal recess obstruction as the cause




CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.